

Unified Lifelong Action Learning: A Real-Video Pipeline for Continual Learning, Knowledge Distillation, and Few-Shot Meta-Adaptation on UCF101

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Abstract—We present *Unified Lifelong Action Learning* (ULAL), an end-to-end pipeline that studies how a single video action recogniser can keep learning new actions over time without forgetting the old ones, trained directly on real video frames. The pipeline is built as five linked studies and a reference study, each answering one question and each comparing several methods under one shared budget. (i) *How do we measure honestly?* UCF101’s shipped split leaks 80% of its source groups across train and test; we replace it with a group-disjoint split that leaks none. (ii) *Which backbone?* A five-way bake-off selects a ResNet50+LSTM teacher (93.9% on the base classes). (iii) *How do we stop forgetting?* On a $10 \rightarrow 15 \rightarrow 20$ class stream, rehearsal keeps 89.8% while naive fine-tuning collapses to 26.0% and the data-free methods (EWC, LwF) trail. (iv) *How do we shrink the model?* Cosine feature distillation gives a MobileNetV3-Small student that keeps 82.7% at $14\times$ fewer parameters; with only 32 training clips per class, a new class-prototype alignment objective raises the student from 61.5% to 67.9%. (v) *How do we learn new classes from a few clips?* A nine-arm comparison with a matched fine-tune control shows MAML+replay is the best-balanced few-shot adapter (harmonic mean 0.60). (vi) *What does a frozen foundation model give?* On frozen CLIP features, a small adapter trained on one clip per class reaches 75.0% on the base classes against 17.9% for our ResNet50+LSTM, and with 16 clips training-free text + visual prototypes reach 90.1%—at the cost of an 88M-parameter encoder, $50\times$ the student. A final capstone chains the winner of every stage into one pipeline and tests it on real YouTube videos of five unseen sports, each learned from 10 clips: after 13 updates the MAML student recognises 0.16 of the clips of unseen test videos where the plain distilled student recognises none, and after 26 updates 0.36, as much as the $14\times$ larger teacher (0.37) and above the plain student (0.23); with 65 updates the plain student catches up and passes it (0.57 against 0.47)—meta-learning pays off exactly when learning must be fast. The whole pipeline runs on a single free Colab T4 GPU.

Index Terms—continual learning, catastrophic forgetting, action recognition, knowledge distillation, meta-learning, few-shot learning, model compression.

I. INTRODUCTION

Deep video models forget. Fine-tuning a trained action recogniser on a set of new classes collapses its accuracy on the classes it already knew—*catastrophic forgetting* [1]. Lifelong,

or continual, learning aims to add classes over time while preserving what was already learned [5]. Video makes this markedly harder than images: a clip is a large spatio-temporal tensor, so storing and replaying past data is expensive, and the temporal model on top of the per-frame features adds a second place for representations to drift.

A realistic deployment needs several abilities at once. It must (a) be measured on data it has never seen, (b) start from strong features, (c) retain old classes as new ones arrive, (d) be small enough to run on a modest device, and (e) still pick up a brand-new class from only a handful of examples. Each of these is usually studied in isolation. We study them *together*, as one pipeline on one dataset, trained *directly on real video frames*, so every reported number reflects the model that would actually be deployed.

The pipeline (Fig. 1) is built on UCF101 [14] as five linked studies (S1–S5), each a notebook answering one question, and a capstone that chains the winner of every study. A sixth study (S6) compares meta-learning with a different route to few-shot learning: a frozen CLIP foundation model. The organising idea is a *moving freeze boundary*: the recogniser is one shared network (a CNN applied to every frame, an LSTM over the frame features, and a classifier head that *grows* as classes arrive), and each stage decides which of its parts may still change.

Contributions.

- **An honest measurement protocol.** We quantify the 80% source-group leakage in UCF101’s shipped split and replace it with a group-disjoint split that leaks nothing.
- **A controlled study per stage:** five backbones, five continual-learning methods, four distillation objectives, and nine meta-adaptation arms, each drawn as its own schematic and compared under one shared budget and seed.
- **A fine-tune control that isolates meta-learning:** the MAML start is compared with an ordinary fine-tune start that saw the *same* extra data.

- **A foundation-model reference:** few-shot learning with a frozen CLIP encoder (text + visual prototypes, linear probe, MLP adapter) against our ResNet50+LSTM trained on the same few clips.
- **A real-video capstone:** the winners of all stages are chained into one pipeline and taught five unseen sports from a few YouTube clips, tested on YouTube videos they never saw.

II. RELATED WORK

Continual learning. Approaches fall into three families. *Regularisation* methods add a penalty that discourages changing parameters important to old tasks: EWC [1] weights the penalty by the Fisher information, and synaptic-intelligence variants use online importance estimates [2]. *Functional* methods keep the model’s outputs stable: Learning without Forgetting (LwF) [3] distills the old model’s own predictions on new data. *Rehearsal* methods store and replay a small set of past exemplars; iCaRL [4] combines rehearsal with distillation. We evaluate a representative of each family on video, where the cost of storing exemplars is a first-order concern. **Knowledge distillation.** A compact student can be trained to match a larger teacher’s softened logits [6] or its intermediate features [7]; we compare logit distillation against two feature objectives. **Meta and few-shot learning.** MAML [8] learns an initialisation from which a few gradient steps adapt to a new task; first-order MAML drops the second-order term for efficiency. Few-shot class-incremental learning (FSCIL) [9] adds new classes from few examples while retaining a large base set, and reports the harmonic mean of base and new accuracy so that neither side can be ignored; we adopt that metric.

III. SYSTEM, NOTATION, AND PROTOCOL

A. The shared recogniser

Every stage uses the same recogniser (Fig. 2). A clip is $T=16$ RGB frames at 224×224 , ImageNet-normalised. A CNN backbone ϕ is applied to each frame independently, producing frame features $\phi(x_1), \dots, \phi(x_T)$. An LSTM reads that sequence and its final hidden state $h_T \in \mathbb{R}^{256}$ is the clip feature. A linear head W maps h_T to class scores:

$$h_T = \text{LSTM}(\phi(x_1), \dots, \phi(x_T)), \quad z = W h_T. \quad (1)$$

The head *grows* as classes arrive: adding k classes appends k rows to W and keeps the existing rows. The teacher uses ResNet50 as ϕ (2048-d per frame), the student MobileNetV3-Small (576-d); both use the same 256-d LSTM, so their clip features h can be compared directly.

B. The moving freeze boundary

The stages differ in *which* parameters they train. Stage 2 trains the whole recogniser. Stage 3 trains the whole recogniser again for each new task (every method in that study trains the same parameters), and the head grows. Stage 4 freezes the teacher and trains the whole student. Stage 5 freezes the lower student backbone and meta-trains only its last four blocks, the LSTM, and the head, so the old features are protected while the

TABLE I
BUDGET PER STAGE (SHARED BY EVERY METHOD COMPARED IN THAT STAGE).

Stage	Key settings
Backbone (S2)	Adam, lr 10^{-4} , 5 epochs
Continual (S3)	Adam, lr 10^{-4} , 5 epochs/task; replay 15/class; LwF weight 3.0, $T=5$; EWC $\lambda=2800$
Distillation (S4)	Adam, lr 10^{-4} , 15 epochs; $T=2$; CE/transfer 0.75/0.25; S4.4: proto 0.5, 3 KMeans prototypes/class; reduced set: 32 clips/class
Meta-train (S5)	5-way 3-shot, 60 episodes $\times$ 8 epochs; 40 sports; inner SGD 5 steps @ 0.05; meta-lr 10^{-4} ; last 4 blocks trained
Adaptation (S5)	8 clips/class, 5 epochs, lr 3×10^{-4} ; replay 8/class; LwF 0.5, $T=2$
CLIP (S6)	ViT-B/32 frozen; $k \in \{1, 2, 4, 8, 16\}$; hybrid 0.4/0.6; probe $C=0.01$; MLP AdamW 10^{-3} , 50 epochs; baseline Adam 10^{-4} , 10 epochs
Capstone	replay 15/class (10 $\rightarrow$ 20 $\rightarrow$ 30); cosine KD, 10 epochs; MAML 5-way 5-shot, 35 sports (1 clip/video); YouTube: 10 clips/sport, 1/2/3/5 epochs, lr 3×10^{-4} , replay 8/class

upper layers reshape. Within a stage every compared method trains the *same* parameters, so only the mechanism under test differs.

C. Datasets and budget

All experiments use UCF101 on one free Colab T4 GPU. The group-disjoint split assigns whole source groups to one split, 18/3/4 groups per class (train/val/test), keeping the usual $\approx 72/12/16\%$ proportions (Section IV). Table I lists the budget of each stage; seeds are fixed and reset before every training run.

D. How to read the numbers

Test accuracy is the fraction correct on the unseen, group-disjoint test set. *Old* and *New* split it into classes learned earlier and the newest task; *Base* is retention on the original classes. When old classes outnumber new ones, a plain average mostly measures the old side and rewards a method that learns nothing new, so for few-shot learning we also report the harmonic mean

$$H = \frac{2 \text{acc}_{\text{new}} \text{acc}_{\text{old}}}{\text{acc}_{\text{new}} + \text{acc}_{\text{old}}}, \quad (2)$$

which is high only when *both* sides are high [9].

IV. STAGE 1 — DATA SPLIT AND LEAKAGE

Why this comes first. A UCF101 filename `v_Class_gXX_cYY.avi` encodes a *group* gXX (one source recording) and a *clip* cYY (one segment of it). All clips of a group share the same actor, background and camera. If the split is made per clip, segments of the *same* recording land in train and test, and the model can recognise the *scene* instead of the *action*. We therefore assign each whole group to a single split (panel S1 of Fig. 1).

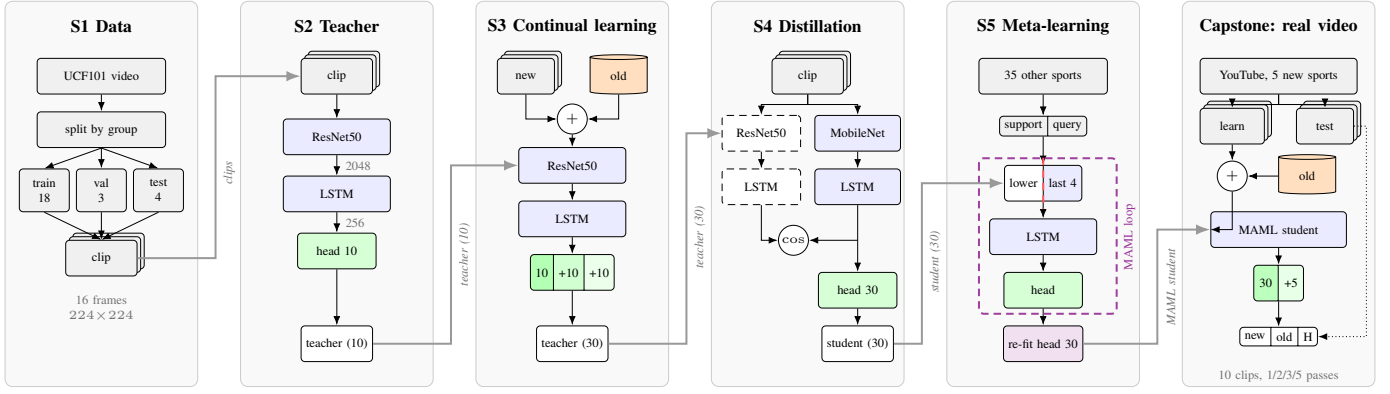


Fig. 1. **The ULAL architecture.** One column per stage, read top to bottom; the grey arrows hand the result of each stage to the part of the next stage it feeds (clips to the teacher, the 10-class teacher to continual learning, the 30-class teacher to distillation, the student to meta-learning, the MAML student to the real-video test). Every column uses the winner of its study (Sections IV–VIII); Section X walks through the whole pipeline. Grey: data; blue: trained; dashed: frozen; orange: stored old clips; violet: meta-learning; green: classifier head; red dashed: freeze boundary.

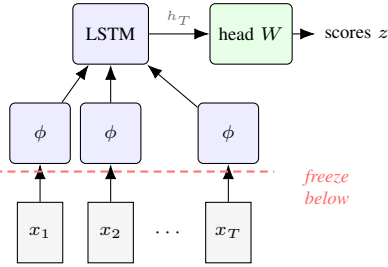


Fig. 2. **The shared recogniser.** The backbone ϕ runs on each of the 16 frames; an LSTM summarises them into a 256-d clip feature h_T ; a growing head classifies. The freeze boundary (dashed) marks how deep training may reach in a stage.

TABLE II

SOURCE GROUPS SHARED BETWEEN SPLITS (20 STUDIED CLASSES, 2,679 VIDEOS).

Split	Groups	Groups in >1 split	Leak
Shipped (per clip)	500	400	80%
Group-disjoint	500	0	0%

Result and reading. Table II: the shipped split puts 400 of 500 groups in more than one split, so 80% of the test recordings were partly seen in training; the group-disjoint split puts none. It keeps the usual 72/12/16% proportions, so every later accuracy is lower than commonly reported on UCF101 only because the inflation is gone—and since the same leak would inflate every later stage, fixing it once at the source makes the whole study trustworthy.

V. STAGE 2 — BACKBONE SELECTION

Design. The backbone ϕ is the only part that changes: each candidate is plugged into the same recogniser (clip $\rightarrow \phi \rightarrow$ LSTM $\rightarrow$ head, panel S2 of Fig. 1) and trained with the same budget and seed. We expect a network with no prior knowledge (a CNN trained from scratch) to struggle on a few thousand clips, and ImageNet-pretrained networks to cluster high.

TABLE III

BACKBONE BAKE-OFF ON THE 10 BASE CLASSES (GROUP-DISJOINT TEST).

Backbone ϕ	Test acc. (%)	Params (M)	ms/clip
ScratchCNN	52.8	103.1	10.2
ResNet18	90.1	12.0	15.7
DenseNet121	92.0	8.3	53.1
VGG19-BN	92.0	20.8	115.7
ResNet50 (selected)	93.9	25.9	54.6

Result and why. Table III. The from-scratch CNN has the most parameters (103M, because its LSTM reads a large flattened feature map) yet the worst accuracy (52.8%): capacity without useful priors does not help on little data. All four pre-trained backbones exceed 90%, because the hard part—good spatial features—is solved before the LSTM sees anything. **ResNet50 is selected:** it is the most accurate (93.9%), its 2048-d feature gives the later stages the richest representation, and at 55 ms per clip it is half as slow as VGG19-BN.

VI. STAGE 3 — CONTINUAL LEARNING

Design. Starting from the 10-class teacher, the head grows and the model learns two new tasks, $10 \rightarrow 15 \rightarrow 20$ classes. The five methods of Fig. 3 train the same network with the same budget; they differ only in *what protects the old classes*. Writing $\mathcal{L}_{CE}$ for cross-entropy on the batch:

- **S3.0 Naive:** $\mathcal{L} = \mathcal{L}_{CE}(\text{new})$ — nothing protects the past.
- **S3.1 EWC** [1]: adds $\frac{\lambda}{2} \sum_i F_i(\theta_i - \theta_i^*)^2$, a penalty that keeps the weights θ important for old classes (Fisher F_i) near their old values θ^* .
- **S3.2 LwF** [3]: adds $\lambda \text{KD}(z_{1:C_{\text{old}}}, z_{1:C_{\text{old}}}^*)$, keeping the old-class scores of the new model close to those of a frozen copy of the old model, computed on the new clips.
- **S3.3 Replay:** stores 15 old clips per class and mixes them 1:1 into every new batch, $\mathcal{L} = \mathcal{L}_{CE}(\text{new} \cup \text{old})$.
- **S3.4 Replay+LwF** [4]: both.

Results. Table IV gives the final accuracies and Table V shows *when* each method forgets.

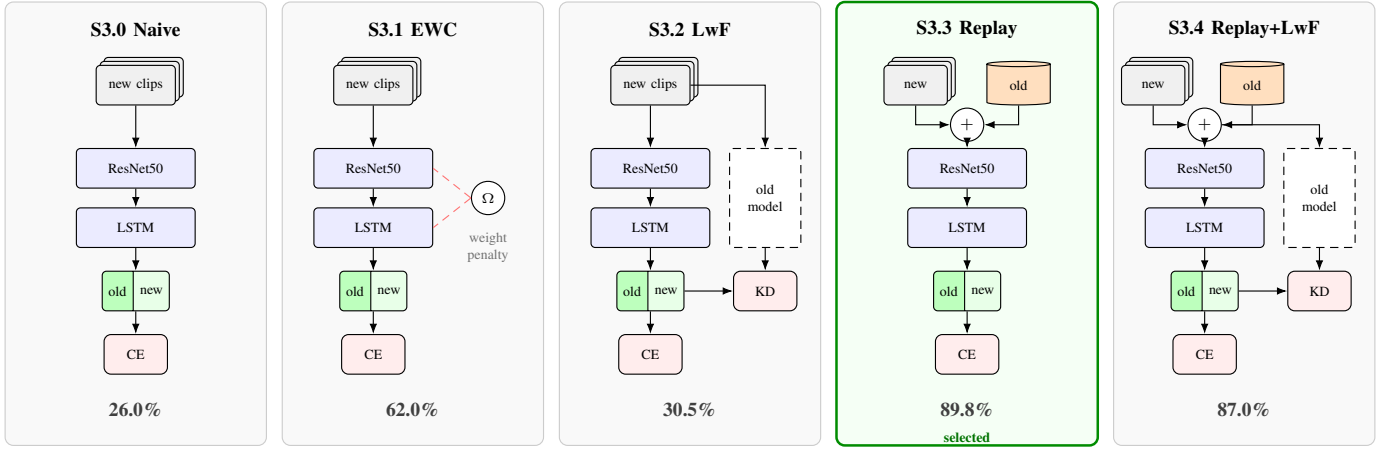


Fig. 3. **Stage 3: the five continual-learning methods**, top to bottom from the input clips to the loss. All train the same network (blue) with a head that grows by the new classes; they differ only in what protects the old classes: nothing (S3.0), a weight penalty $\Omega = \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2$ (S3.1), the scores of a frozen copy of the old model (S3.2, dashed), stored old clips (S3.3, orange), or both (S3.4). Bottom: final accuracy on all 20 classes (Table IV).

TABLE IV
CONTINUAL LEARNING AFTER THE FULL 10 $\rightarrow$ 15 $\rightarrow$ 20 STREAM (%).
FINAL: ALL 20 CLASSES; OLD: CLASSES 0–14; NEW: LAST TASK 15–19;
BASE: ORIGINAL 10.

Method	Final	Old (0–14)	New (15–19)	Base (0–9)
S3.0 Naive	26.0	0.9	94.9	0.5
S3.2 LwF	30.5	5.3	100.0	0.0
S3.1 EWC	62.0	71.1	36.8	76.9
S3.4 Replay+LwF	87.0	82.6	99.2	83.0
S3.3 Replay	89.8	86.7	98.3	89.2

TABLE V
WHEN EACH METHOD FORGETS: ACCURACY PER TASK (%) AFTER
LEARNING TASK 1 AND AFTER LEARNING TASK 2. BEFORE ANY NEW
TASK THE BASE ACCURACY IS 93.9.

Method	after task 1		after task 2		
	Base	Task 1	Base	Task 1	Task 2
S3.0 Naive	22.6	98.2	0.5	1.8	94.9
S3.1 EWC	80.7	79.1	76.9	60.0	36.8
S3.2 LwF	26.4	97.3	0.0	15.5	100.0
S3.3 Replay	89.6	94.6	89.2	81.8	98.3
S3.4 Replay+LwF	87.3	96.4	83.0	81.8	99.2

Why each method behaves as it does. *S3.0 Naive* learns every new task almost perfectly (98.2% on task 1, 94.9% on task 2) but the base falls from 93.9% to 22.6% after one task and to 0.5% after two: with only new labels in every batch, the shared weights are rewritten for the newest task. *S3.2 LwF* forgets almost as much (base 26.4% $\rightarrow$ 0.0%) even though it protects the old scores. The reason is the setting: the original LwF [3] gives every task its own head and is tested with the task known, whereas here one head covers all classes and the task is not known at test time. The distillation term only keeps the *ranking among old classes*, while the cross-entropy on new clips keeps raising the new classes above all old ones; the old clips are then out-voted by the newest classes—in

the confusion matrices almost every forgotten base clip is predicted as a *new* class, not as another old one. *S3.1 EWC* fails the other way: its penalty keeps the base (76.9%) but also blocks the new task (36.8% on task 2), a stability/plasticity lock. *S3.3 Replay* is the only method that keeps every task high (base 89.2%, task 1 81.8%, task 2 98.3%): the stored clips come with their old labels, so every batch also tells the model what the old classes look like and puts them back in the competition. *S3.4 Replay+LwF* is close (87.0%) but slightly lower: the buffer already protects the old classes, and the extra distillation term only slows the new task. **Choice for the pipeline: replay** (panel S3 of Fig. 1). On video, storing a few clips beats every data-free method.

VII. STAGE 4 — KNOWLEDGE DISTILLATION

Design. The 25.9M-parameter teacher is too big for a small device, so we train a MobileNetV3-Small+LSTM student (≈ 1.8 M parameters, $14\times$ smaller). The five objectives of Fig. 4 share the ImageNet-pretrained student, data, budget and seed, and all use $\mathcal{L} = 0.75 \mathcal{L}_{\text{CE}} + 0.25 \mathcal{L}_{\text{transfer}}$:

- **S4.0 CE only:** $\mathcal{L}_{\text{transfer}} = 0$ (no teacher, the baseline).
- **S4.1 KD** [6]: match the teacher’s softened scores, $T^2 \text{KL}(\sigma(z_t/T) \parallel \sigma(z_s/T))$, $T = 2$.
- **S4.2 Cosine:** align the 256-d clip features by direction, $1 - \cos(h_s, h_t)$.
- **S4.3 MSE** [7]: a learnable regressor g maps h_s onto the teacher’s feature before an ℓ_2 match, $\|g(h_s) - h_t\|_2^2$ (g is dropped at inference).
- **S4.4 Proto-align** (new): before training, the teacher’s clip features of every class are clustered with KMeans into 3 *prototypes*; during training each student feature is pulled to the nearest prototype of its own class, $\min_j \|h_s - \mu_{y,j}\|_2^2$, on top of CE and KD: $\mathcal{L} = 0.75 \mathcal{L}_{\text{CE}} + 0.25 \text{KD} + 0.5 \mathcal{L}_{\text{proto}}$. The target is a class-level summary of the teacher (in the spirit of prototypical networks [13]) rather than the teacher’s feature of each single clip.

TABLE VI

DISTILLATION INTO THE MOBILENETV3-SMALL STUDENT (20-CLASS GROUP-DISJOINT TEST, %). SAME PRETRAINED STUDENT, BUDGET, SEED AND INITIALISATION; ONLY THE TRANSFER TERM DIFFERS. FULL: ALL 1,917 TRAINING CLIPS (≈ 96 PER CLASS); 32/CLASS: 640 TRAINING CLIPS.

Model	Full	32/class	Params (M)
Teacher (ResNet50, Replay+LwF)	87.0	86.8	25.9
S4.0 Student, CE only	80.6	61.5	1.8
S4.1 Student, CE + KD	79.7	62.0	1.8
S4.2 Student, CE + cosine	82.7	59.7	1.8
S4.3 Student, CE + MSE (g)	75.9	61.3	1.8
S4.4 Student, proto-align (new)	TODO	67.9	1.8

Because a pretrained student already learns most of what it needs from the labels, we also repeat the study with a *reduced training set*: 32 clips per class (640 clips, about one third of the full 1,917), with the same test set. The teacher is the same Stage-3 model, trained on the full set, so it knows more than the student sees—the situation in which distillation should help most. Fig. 5 shows proto-align in detail.

Results and why. Table VI. The first lesson is *compression*: a student $14\times$ smaller stays within 4.3 points of its teacher (82.7% vs 87.0%), because the teacher’s decision function transfers. The second is the ranking of the objectives, which follows from what each one asks of the student. *S4.0 CE only* is already strong (80.6%); the student backbone is ImageNet-pretrained, so labels alone carry it far. *S4.1 KD* does not help here (79.7%); the softened teacher scores add little beyond the labels once the student is pretrained. *S4.2 Cosine* is best (82.7%, +2.1 over CE); it asks only that the student’s 256-d clip feature *point in the same direction* as the teacher’s, a gentle target that transfers the teacher’s temporal summary without forcing its scale. *S4.3 MSE* is worst (75.9%); it asks the student to match direction *and* magnitude across two very different backbones, a target that fights the label loss. The gap between the objectives is modest (single seed, 439 test clips), so the headline of this stage is compression, with cosine as the best transfer term. **Choice for the pipeline: cosine distillation** (panel S4 of Fig. 1).

With few training clips: proto-align wins. With 32 clips per class every student loses about 19 points (CE: 80.6% $\rightarrow$ 61.5%) and overfits: training accuracy reaches 97–99% while test accuracy stays near 60%. This is where the teacher—trained on the full set—should help, and the right half of Table VI shows which kind of help works. *Per-clip targets add little*: KD (62.0%) and MSE (61.3%) stay at the level of CE, and cosine (59.7%) is even slightly below it. They copy what the teacher says about *each* training clip; with only 32 clips per class these are 32 more targets to fit, not new information about the class. *Proto-align is the clear winner* at 67.9%, +6.4 points over CE and +5.9 over KD, well above the noise of a single run (one standard error is about 2.3 points on 439 test clips); it also narrows the gap to the teacher from 25.3 to 18.9 points. Its target is different in kind: a *class-level* summary of how the teacher sees each class, built from all

TABLE VII

FEW-SHOT META-ADAPTATION: 20 BASE + 5 NEW SPORTS, 8 CLIPS PER NEW SPORT, IDENTICAL ADAPTATION FOR EVERY ARM, RANKED BY THE HARMONIC MEAN H (EQ. 2).

Start	Protection	New	Base	H
S5.2 MAML	replay	0.625	0.576	0.600
S5.2 MAML	replay+LwF	0.580	0.542	0.561
S5.1 Fine-tune	replay	0.464	0.617	0.530
S5.1 Fine-tune	replay+LwF	0.446	0.626	0.521
S5.0 Plain	replay	0.295	0.677	0.411
S5.0 Plain	replay+LwF	0.223	0.724	0.341
S5.0 Plain	none	0.527	0.189	0.278
S5.2 MAML	none	0.661	0.011	0.022
S5.1 Fine-tune	none	0.500	0.011	0.022

its clips. Pulling every student feature to the nearest of three class prototypes makes the clips of a class gather around a few centres—exactly the regularisation an overfitting student needs—while three prototypes instead of one still leave room for different ways of performing the same action. In short, when data is scarce the teacher helps most by showing the student *where each class lives* in feature space, not what each single clip looks like. With the full training set cosine remains the measured best and stays in the pipeline; **TODO** proto-align with the full training set—if it also beats cosine there, it should replace cosine in panel S4 of Fig. 1.

VIII. STAGE 5 — FEW-SHOT META-ADAPTATION

Design. Given the 20-class student, how well can it learn *five brand-new sports* from only a few clips? Every arm uses the same adaptation—the same 8 clips per new sport, 5 epochs, learning rate and seed, a head grown from 20 to 25 classes—so only two things vary (Fig. 6): the *starting point* and the *protection* of the old classes. The three starting points are:

- **S5.0 Plain:** the distilled student as it is.
- **S5.1 Fine-tune:** the student after ordinary supervised training on 40 other sports. This is the control: it sees the same extra data as MAML.
- **S5.2 MAML** [8]: the student after first-order MAML on the same 40 sports. Each episode draws 5 sports with 3 support and 3 query clips; the *inner* loop adapts a copy on the support clips, $\theta' = \theta - \alpha \nabla_{\theta} \mathcal{L}^{\text{sup}}(\theta)$ (5 SGD steps), and the *outer* loop moves the start by the loss of the adapted copy on the query clips, $\theta \leftarrow \theta - \beta \nabla_{\theta'} \mathcal{L}^{\text{qry}}(\theta')$.

For S5.1 and S5.2 only the last four backbone blocks, the LSTM and the head are trained, and a fresh 20-class head is then re-fitted to the new features (a short linear probe on 12 clips per class). Each start is adapted with no protection, with replay (8 old clips per class), or with replay+LwF, giving nine arms ranked by the harmonic mean of Eq. (2).

Results. Table VII ranks the nine arms; Table VIII shows how the new-sport accuracy grows with the number of clips. Before adaptation the student scores 75.9% on the base classes; during meta-training the query accuracy rises from 0.83 to 0.98 over 8 epochs, so the start is really being reshaped.

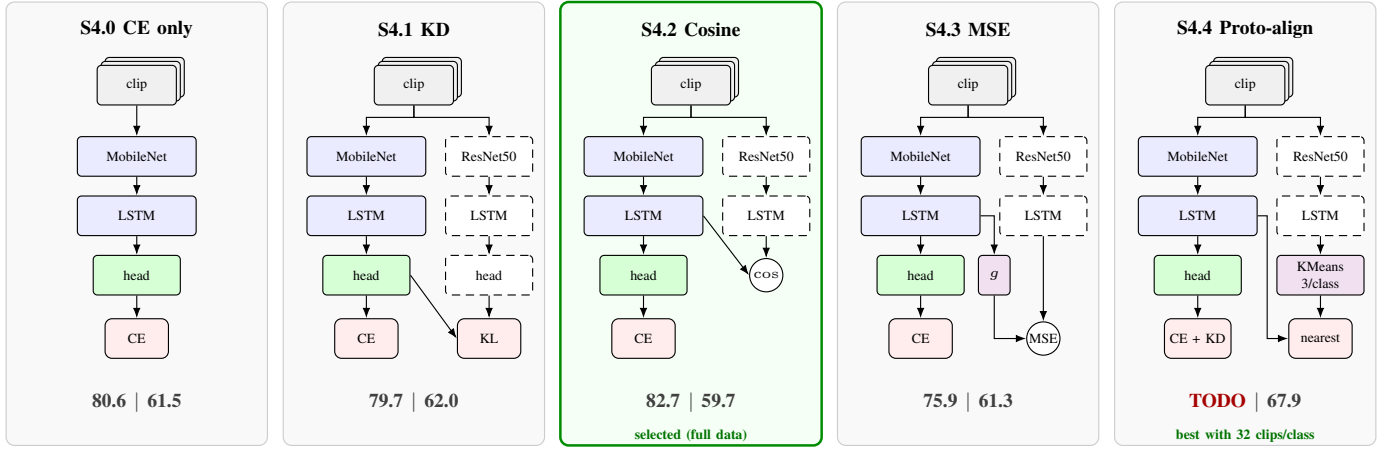


Fig. 4. **Stage 4: the five distillation objectives**, top to bottom. The pretrained student (blue, left of each panel) always learns from the labels (CE); the frozen teacher (dashed, right) adds nothing (S4.0), its softened scores (S4.1), its clip feature by direction (S4.2), its clip feature through a learnable regressor g (S4.3), or KD plus the nearest of 3 KMeans prototypes of the student’s class (S4.4, new; detail in Fig. 5). Bottom: student test accuracy (%) with the full training set | with 32 clips per class (Table VI).

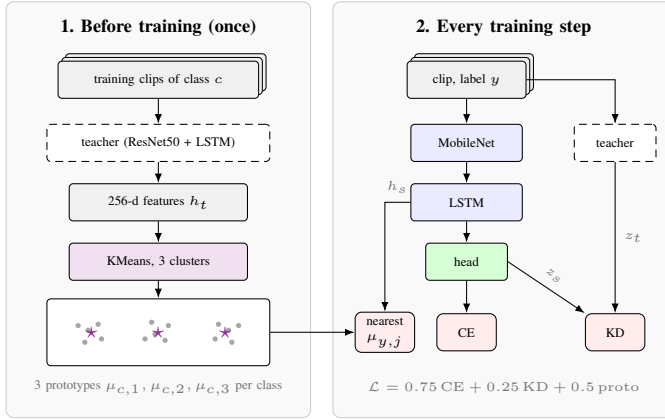


Fig. 5. **S4.4 Proto-align in detail**. (1) Once, before training, the frozen teacher encodes the training clips of every class and KMeans groups their 256-d features into 3 prototypes per class (stars). (2) At every step the student learns from the label (CE), from the teacher’s softened scores (KD), and from a prototype term that pulls its clip feature h_s to the nearest prototype of its own class y , $\min_j \|h_s - \mu_{y,j}\|^2$. The teacher and the prototypes are never updated; only the student (blue) trains.

learns the new sports best (0.661), because its start is built to move quickly onto a new task—and with nothing holding it back it moves all the way. *Replay* fixes the base in every case (0.576 to 0.724) but slows new learning, most for the plain start (0.295). *MAML+replay* combines a start that learns fast with stored clips that hold the old classes, and is the best balance (H 0.600). The key comparison is with the control: *fine-tune+replay* saw the same 40 sports and reaches only 0.530, so the extra 0.07 comes from meta-learning, not from the extra data. The LwF variants are always slightly worse: with only a few new clips, keeping the old scores fixed also holds back the fresh new-class rows. Table VIII shows that MAML’s advantage is a property of the *start*: it is largest with 1 clip (0.26 against 0.00–0.03) and disappears by 10 clips (0.65 against 0.66 for the plain start), when there are enough steps for any start to learn. **Choice for the pipeline: MAML meta-training, adapted with replay** (panel S5 of Fig. 1).

IX. STAGE 6 — FEW-SHOT LEARNING WITH A FROZEN CLIP ENCODER

The question. Stage 5 teaches our own small student to learn fast. A different route is to borrow a *foundation model* that already knows a very wide range of images and keep it frozen: CLIP [12] (ViT-B/32) was trained on 400M image–text pairs. Can it learn our actions from a handful of clips, or even from their names alone? This study is a reference for Stage 5; it is not part of the pipeline of Fig. 1.

Design (Fig. 7). Every one of the 16 frames goes through the frozen CLIP image encoder; the 16 frame embeddings are averaged into one normalised 512-d clip feature f . The task is the 10 base classes, learned from $k \in \{1, 2, 4, 8, 16\}$ training clips per class and tested on the 212 clips of the group-disjoint test set (one clip is 0.47 points; chance is 10%). The k clips are the first k of each class in group order, so for $k \leq 4$ they all come from one source recording, for $k = 8$ from two and for $k = 16$ from three or four. Four methods are compared:

TABLE VIII

NEW-SPORT ACCURACY VERSUS THE NUMBER OF CLIPS PER NEW SPORT.

Start	Protection	1	3	5	10
S5.0 Plain	none	0.00	0.13	0.43	0.66
S5.1 Fine-tune	none	0.03	0.37	0.54	0.59
S5.2 MAML	none	0.26	0.42	0.55	0.65
S5.0 Plain	replay	0.00	0.04	0.23	0.45
S5.1 Fine-tune	replay	0.00	0.22	0.34	0.49
S5.2 MAML	replay	0.16	0.19	0.44	0.55

Why each arm behaves as it does. Without protection every start learns the new sports but wipes the base (base 0.011–0.189): with only new clips, the whole network is rewritten, exactly as in naive continual learning. *MAML alone*

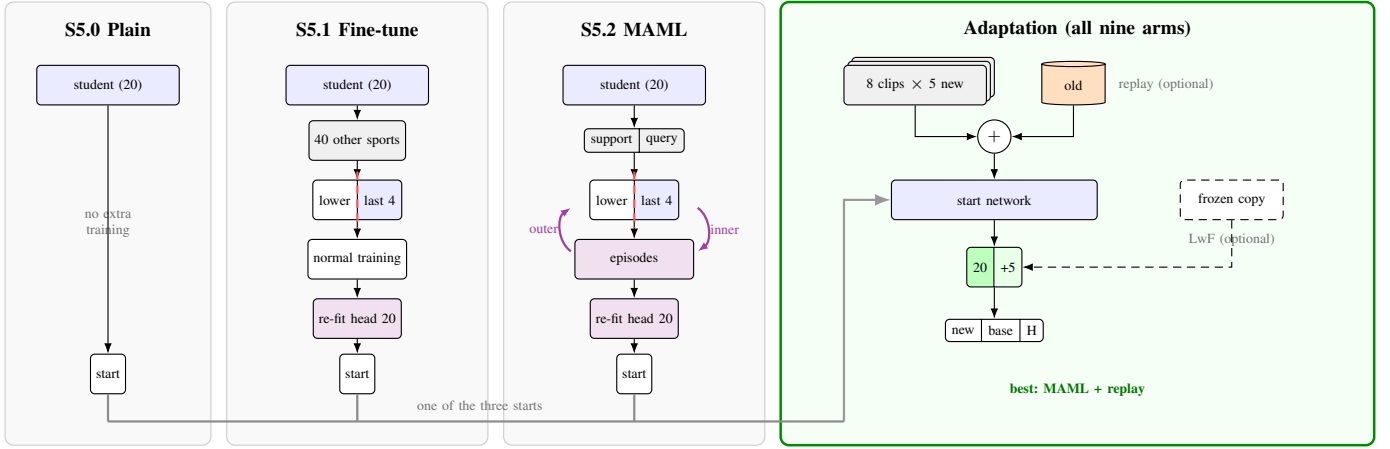


Fig. 6. **Stage 5: three starting points and one shared adaptation**, top to bottom. The plain student (S5.0), the student after normal training on 40 other sports (S5.1, the control), and the student after MAML on the same sports (S5.2, violet: inner/outer loop) are each adapted in the same way: 8 clips of 5 new sports, a head grown from 20 to 25 classes, with optional replay (orange) and optional LwF (dashed). Red dashed: only the last 4 blocks, LSTM and head are trained before adaptation.

TABLE IX

STAGE 6: TEST ACCURACY (%) ON THE 10 BASE CLASSES (212 TEST CLIPS, CHANCE 10%) WITH k TRAINING CLIPS PER CLASS. “TRAINED” COUNTS THE TRAINABLE PARAMETERS; THE CLIP IMAGE ENCODER (≈ 88 M) IS FROZEN. —: NOT RUN.

Method	Trained	$k=1$	2	4	8	16
S6.0 ResNet50+LSTM	2.4M	17.9	15.1	36.3	50.9	61.3
S6.1 Hybrid proto.	0	—	—	—	—	90.1
S6.2 Linear probe	5k	—	—	—	—	89.6
S6.3 MLP adapter	0.13M	75.0	73.1	74.1	79.2	86.8

- **S6.0 Baseline:** our ResNet50+LSTM with the ImageNet backbone frozen and the LSTM and head trained on the same k clips (Adam 10^{-4} , 10 epochs). It is not the Stage-2 teacher: it sees only the k clips.
- **S6.1 Hybrid prototypes:** each class name becomes the prompt “a video of a person performing $\langle \text{class} \rangle$ ”, encoded by the CLIP text encoder into a text prototype t_c ; it is mixed with the mean feature m_c of the k clips of the class, $p_c = 0.4t_c + 0.6m_c$ (normalised), and a clip gets the class with the highest $\cos(f, p_c)$. Nothing is trained.
- **S6.2 Linear probe:** logistic regression on f ($C = 0.01$), trained on the k clips per class.
- **S6.3 MLP adapter:** a small MLP (512 $\rightarrow$ 256, batch norm, ReLU, dropout 0.3, then the classes) on f , trained on the k clips per class (AdamW 10^{-3} , weight decay 10^{-2} , 50 epochs).

S6.0 and S6.3 are run for every k ; S6.1 and S6.2 only for $k = 16$.

Results (Table IX). *With CLIP features one clip is enough to start.* From a single clip per class the MLP adapter reaches 75.0%, against 17.9% for our ResNet50+LSTM trained on the same clip. The baseline has to learn a 2.4M-parameter LSTM from 10–40 clips: up to $k = 4$ its validation loss stays above 2.0, close to a uniform guess ($\ln 10 \approx 2.3$), and with 16 clips

it reaches only 61.3%, 25.5 points below the adapter. The two are not trained for the same number of steps (10 epochs against 50), so at small k part of the gap is under-training; at $k = 16$, where the baseline fits all its training clips, the gap comes from the features.

More clips help only when they come from new recordings. The adapter is flat from 1 to 4 clips (75.0, 73.1, 74.1%, a spread of 4 test clips): these clips all come from the same recording, so they add almost no new information. It rises once the clips come from new recordings: 79.2% with 8 (two recordings) and 86.8% with 16 (three or four).

With 16 clips the simplest classifiers win. The hybrid prototypes, which train nothing, reach 90.1% and the linear probe 89.6%, both above the MLP adapter (86.8%) and only about 4 points below the Stage-2 teacher trained end-to-end on all the training clips (93.9%). On 160 clips the 0.13M-parameter MLP overfits more than a 5k-parameter probe or a class mean.

MAML (Stage 5) or CLIP (Stage 6)? The two answer the same need—learning a class from a few clips—in opposite ways, and each wins on different criteria:

- **Prior knowledge.** CLIP brings what it learned from 400M web images and captions, which is why one clip per class already gives 75.0% here. MAML only knows the UCF101 sports it was meta-trained on, and must still update the network.
- **Forgetting.** With frozen CLIP features, adding a class means adding a prototype (or a row of a small classifier); the old classes are untouched, so nothing is forgotten and no replay buffer is needed. MAML updates the network and needs replay to keep the old classes.
- **Size.** The frozen CLIP image encoder has ≈ 88 M parameters, about $50\times$ the 1.8M distilled student and more than the 25.9M teacher: it gives up the compression that Stage 4 was built for.
- **Time.** Averaging frame features ignores their order, while

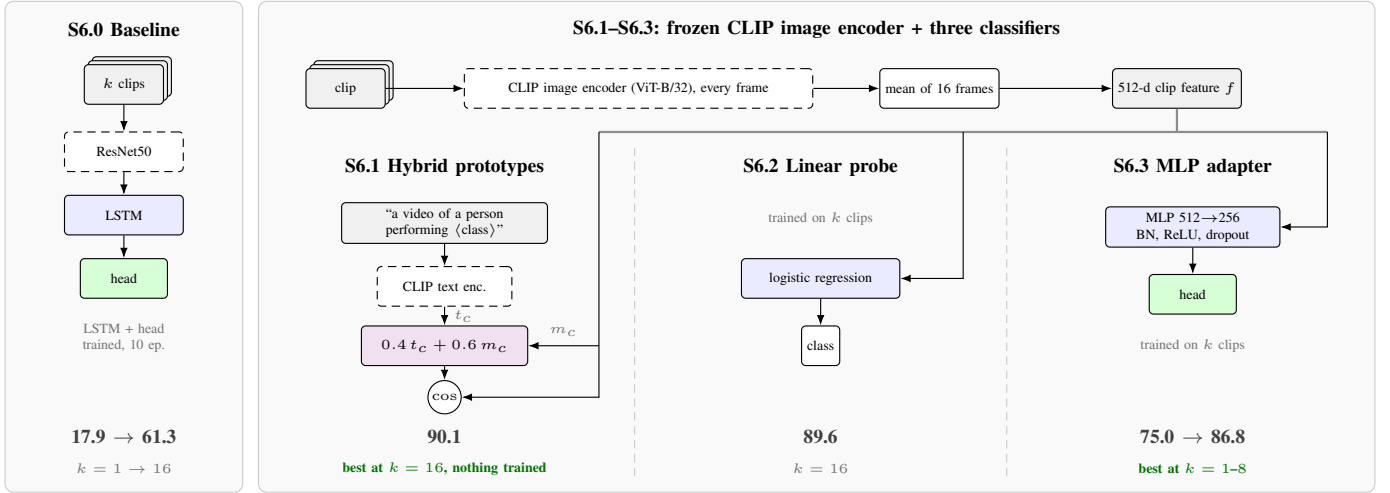


Fig. 7. **Stage 6: few-shot learning with a frozen CLIP encoder**, top to bottom. Left: our ResNet50+LSTM, its LSTM and head trained on the same k clips. Right: the frozen CLIP image encoder (dashed) turns every clip into one 512-d feature f (average of its 16 frames), and three classifiers use it: text + visual prototypes (S6.1; t_c from the class name, m_c the mean feature of the k clips of class c), a linear probe (S6.2), and a small MLP adapter (S6.3). Blue: trained on the k clips. Bottom: test accuracy (%), from 1 to 16 clips per class where the whole sweep was run.

our recogniser has an LSTM. The 10 base classes can largely be told apart from the scene and objects of a single frame (a horse, a pommel horse, a tabla), which suits CLIP; actions that differ mainly in their motion would test this limit.

The two studies do not share a task (Stage 5 adds 5 new sports to 20 known classes and is scored with the harmonic mean; Stage 6 learns 10 classes from nothing), so their numbers cannot be compared directly. A like-for-like answer needs the same task for both: **TODO** evaluate S6.1 and S6.3 on the YouTube test of Section X, next to the MAML student.

X. THE ULAL PIPELINE ON REAL VIDEO

Building the pipeline from the winners. The final notebook chains the selected method of every stage into one lifelong model, following Fig. 1 from top to bottom, now on a longer $10 \rightarrow 20 \rightarrow 30$ stream:

- 1) **S1 Data:** all clips come from the group-disjoint split.
- 2) **S2 Teacher:** the ResNet50+LSTM teacher, trained on the 10 base classes.
- 3) **S3 Replay:** the teacher’s head grows to 20 and then 30 classes; every batch mixes new clips with stored old clips (15 per class).
- 4) **S4 Cosine distillation:** the frozen 30-class teacher is distilled into the MobileNet student with CE + cosine.
- 5) **S5 MAML:** the student is meta-trained on 35 other sports—the 40 meta sports of Stage 5 minus the 5 that are among the 30 known classes—(5-way 5-shot, only its last 4 blocks, LSTM and head); each meta clip comes from a different source video, so the query clips are always from other videos than the support clips. Its 30-class head is then re-fitted on 12 clips per class.
- 6) **Real-video test:** 5 sports that no model has seen (golf swing, front crawl, hammer throw, handstand push-ups,

TABLE X
THE PIPELINE ON UCF101 (GROUP-DISJOINT TEST).

Stage	Model	Classes	Acc. (%)	Params
S2	ResNet50 teacher	10	93.9	25.9M
S3	teacher + replay	30	79.7	25.9M
	old / new classes		78.8 / 81.4	
S4	student, cosine KD	30	67.2	1.8M
S5	MAML student (re-fitted)	30	60.8	1.8M

handstand walking) are taken from YouTube, 4 videos each, cut into 10 clips per video: 2 videos to learn from and 2 kept only for testing (one front-crawl video could not be downloaded, so front crawl learns from a single video). Each model grows its head from 30 to 35 classes and learns from the same 10 clips per sport, with replay (8 old clips per class), in 1, 2, 3 or 5 passes over the clips (13, 26, 39 or 65 updates): few passes mean that learning must be fast, the case MAML is built for. We adapt the MAML student and, as references, the teacher and the plain distilled student, with the same clips, seed and budget. We report the accuracy on the unseen test videos (new), the accuracy kept on the 30 UCF classes (old), and their harmonic mean.

Results on UCF101. Table X. On the longer stream replay still keeps the model balanced: after two tasks of ten classes each the teacher scores 79.7% on all 30 classes, with old (78.8%) and new classes (81.4%) close together—no collapse towards the newest task. The distilled student keeps 67.2% at $14\times$ fewer parameters; the gap to the teacher is larger than in Stage 4 (30 classes instead of 20, and 10 instead of 15 epochs). MAML meta-training costs the student 6.4 points on the known classes (60.8%): moving the last blocks towards fast adaptation, and re-fitting the head on only 12 clips per class,

TABLE XI

LEARNING 5 UNSEEN SPORTS FROM 10 YOUTUBE CLIPS PER SPORT (WITH REPLAY), BY NUMBER OF PASSES OVER THE CLIPS (UPDATES IN BRACKETS). NEW: ACCURACY ON THE 100 CLIPS OF THE 10 UNSEEN YOUTUBE TEST VIDEOS; OLD: ACCURACY KEPT ON THE 30 UCF CLASSES (BEFORE ADAPTATION: TEACHER 0.797, KD STUDENT 0.672, MAML STUDENT 0.608); H: HARMONIC MEAN. BOLD: BEST PER ROW. THE TEACHER HAS 25.9M PARAMETERS, EACH STUDENT 1.8M.

Epochs	Teacher			KD student			MAML student		
	New	Old	H	New	Old	H	New	Old	H
1 (13)	0.21	0.768	0.330	0.00	0.583	0.000	0.16	0.545	0.247
2 (26)	0.37	0.773	0.500	0.23	0.644	0.339	0.36	0.544	0.433
3 (39)	0.49	0.744	0.591	0.42	0.614	0.499	0.47	0.533	0.499
5 (65)	0.35	0.776	0.482	0.57	0.586	0.578	0.47	0.542	0.503

trades some old-class accuracy for plasticity, as in Stage 5.

Results on real video and why (Table XI). *Few updates: the MAML student is the faster of the two students.* After one pass (13 updates) the plain KD student has learned nothing of the new sports (0.00), while the MAML student already recognises 0.16 of the unseen test clips. After two passes (26 updates) it reaches 0.36 against 0.23, as much as the 14× larger teacher (0.37). This is the ability MAML was selected for in Stage 5: a starting point that moves towards a new class in a few steps.

More updates: the plain student catches up and passes. At three passes the three models are within 0.07 of each other (0.42–0.49), less than one test video. At five passes the plain student is best both on the new sports (0.57) and on the balance (H 0.578), while the MAML student stays at 0.47. The same trend appeared in pilot runs and, as the number of clips grows, in Stage 5 (Table VIII, without protection): meta-learning buys speed, not a higher final accuracy.

Old classes. Replay keeps every model close to its starting level on the 30 UCF classes: the teacher stays at 0.74–0.78 (from 0.80), the KD student at 0.58–0.64 (from 0.67), and the MAML student at 0.53–0.55 (from 0.61). The MAML student has the lowest old accuracy throughout because it starts lower (meta-training cost it 6.4 points, Table X). Its balance is still better than the plain student’s with one or two passes (H 0.247 and 0.433 against 0.000 and 0.339), equal at three and lower at five.

Teacher. The large teacher keeps the old classes best and has the best balance up to three passes (H 0.591). At five passes its new-sport accuracy falls from 0.49 to 0.35 while its old accuracy holds. We read this as over-fitting to the two learning videos of each sport, but the drop is only slightly larger than one test video.

Three limits hold for any reading of this test. There are only 10 test videos (100 clips), so one video changes the new-sport accuracy by up to 0.10 and smaller differences are within noise. One front-crawl video could not be downloaded, so front crawl is learned from a single video. And some sports change setting between videos (the handstand push-up learning videos are filmed in a studio, the test videos in a gym).

XI. DISCUSSION AND LIMITATIONS

Four themes recur across the pipeline. *Measurement matters:* without the group-disjoint split every number would be inflated by scene memorisation, and without the harmonic mean the few-shot ranking would reward models that learn nothing new. *Data beats cleverness for retention:* on video, rehearsal of a few clips per class outperforms every data-free method (EWC, LwF); LwF in particular is designed for a setting where the task is known at test time and collapses when one head must choose among all classes. *Each tool has a regime:* distillation’s win is compression—and with scarce data, a teacher helps most through class-level prototypes rather than per-clip targets—and meta-learning’s win is learning from very few clips in very few updates—with more clips or more updates the simpler student catches up, both in Stage 5 and on real video. *Prior knowledge is the strongest few-shot tool, but it costs size:* frozen CLIP features learn the base classes from one clip (75.0%) where our recogniser needs far more, yet the encoder is 50× the student, so it answers a different deployment (no size limit) than the one ULAL targets.

Limitations. All results use a single seed and one dataset, with short streams (up to 30 classes). Rehearsal stores real clips, so its memory cost grows with the number of classes. The real-video test is small (5 sports, 10 test videos) and depends on downloadable YouTube videos (one of the 20 could not be downloaded); a larger set of videos and several seeds would make its differences at three and five passes conclusive.

XII. CONCLUSION

ULAL is a compact, honest, real-video pipeline for lifelong action recognition on a single free GPU. A leakage-free split makes the numbers trustworthy; a ResNet50+LSTM is a strong teacher (93.9%); rehearsal is the strongest continual learner (89.8% against 26.0% for naive fine-tuning); cosine distillation gives a 14× smaller student that keeps 82.7%, and with scarce data the new prototype alignment is the best way to use the teacher (+6.4 points); and, scored with a harmonic metric and a fine-tune control, MAML+replay is the best-balanced few-shot adapter (H 0.60). As a reference, a frozen CLIP encoder learns the base classes from a single clip (75.0%, 90.1% with 16 clips and no training), the best choice when model size does not matter. Chained into one pipeline and tested on unseen YouTube videos, the meta-trained student is the fastest learner of the two students: after 26 updates it matches the 14× larger teacher on the new sports (0.36 against 0.37) and beats the plain student (0.23), which catches up only with more updates. Each chosen tool adds the ability it was selected for.

AUTHOR CONTRIBUTIONS

The work divides along the pipeline of Fig. 1, and Table XII indexes which author led which part. Each author owns a contiguous span of the pipeline together with a share of the shared infrastructure (the recogniser, the training harness, and the plotting and metrics code), and the two comparison-heavy stages (continual learning and meta-adaptation) sit with

different authors. All three authors jointly designed the study, reviewed each other’s stages for fairness, and wrote the paper.

TABLE XII
AUTHOR CONTRIBUTION INDEX, ORGANISED BY PIPELINE STAGE.

Contributor	Responsibilities
Second B. Author	S1 data split and leakage audit; S2 backbone bake-off; the shared recogniser and training harness.
Third C. Author	S3 continual learning (all five methods, forgetting timelines); S4 knowledge distillation (four objectives, compression analysis).
First A. Author	S5 few-shot meta-adaptation (MAML, fine-tune control, harmonic metric); the ULAL capstone and real-video test; the evaluation protocol and metrics.

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